RAG INTERNSHIP PROJECT

Design & Build a RAG-Based Customer Support Assistant (with LangGraph & HITL)

1. Project Objective

Design and implement a Retrieval-Augmented Generation (RAG) system that:

- Processes a PDF knowledge base
- Retrieves relevant information using embeddings
- Answers user queries contextually
- Uses a graph-based workflow for control logic
- Routes responses based on intent
- Supports Human-in-the-Loop (HITL) escalation

This project emphasizes system design thinking + applied implementation.

2. Mandatory Concepts to Be Applied

1. What is RAG
2. Load a PDF → Chunk → Store embeddings in ChromaDB
3. Query system to retrieve answers from document
4. Graph-based workflow using LangGraph
5. 2-node flow: Input → Process → Output
6. Conditional routing based on intent
7. Customer Support Bot use case
8. Human-in-the-Loop (HITL) escalation

3. Deliverable 1: High-Level Design (HLD)

Objective

Define the overall system architecture.

What your HLD must include:

1. System Overview

- Problem definition
- Scope of the system

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2. Architecture Diagram (Mandatory)

- User interface (CLI / Web)
- Document ingestion pipeline
- Embedding system
- Vector database
- Retrieval layer
- LLM processing layer
- Workflow orchestration (LangGraph)
- HITL system

3. Component Description

- Document Loader
- Chunking Strategy
- Embedding Model
- Vector Store
- Retriever
- LLM
- Graph Workflow Engine
- Routing Layer
- HITL Module

4. Data Flow

- How data moves from PDF → Answer
- Query lifecycle

5. Technology Choices

- Why ChromaDB
- Why LangGraph
- LLM choice
- Any additional tools

6. Scalability Considerations

- Handling large documents
- Increasing query load
- Latency concerns

4. Deliverable 2: Low-Level Design (LLD)

Objective

Define how the system will be implemented internally.

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What your LLD must include:

1. Module-Level Design

- Document Processing Module
- Chunking Module
- Embedding Module
- Vector Storage Module
- Retrieval Module
- Query Processing Module
- Graph Execution Module
- HITL Module

2. Data Structures

- Document representation
- Chunk format
- Embedding structure
- Query-response schema
- State object for graph

3. Workflow Design (LangGraph)

- Nodes: Processing Node, Output Node
- Edges: Flow between nodes
- State: Data flowing between nodes

4. Conditional Routing Logic

- Answer generation criteria
- Escalation criteria
- Low confidence
- Missing context
- Complex query

5. HITL Design

- When escalation is triggered
- What happens after escalation
- How human response is integrated

6. API / Interface Design

- Input format
- Output format
- Interaction flow

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7. Error Handling

- Missing data
- No relevant chunks found
- LLM failure

5. Deliverable 3: Technical Documentation

Objective

Provide a complete explanation of your system as if presenting to engineers.

Your document must include:

1. Introduction

- What is RAG
- Why it is needed
- Use case overview

2. System Architecture Explanation

- Detailed explanation of HLD
- Component interactions

3. Design Decisions

- Chunk size choice
- Embedding strategy
- Retrieval approach
- Prompt design logic

4. Workflow Explanation

- LangGraph usage
- Node responsibilities
- State transitions

5. Conditional Logic

- Intent detection
- Routing decisions

6. HITL Implementation

- Role of human intervention
- Benefits and limitations

7. Challenges & Trade-offs

- Retrieval accuracy vs speed

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- Chunk size vs context quality
- Cost vs performance

8. Testing Strategy

- Testing approach
- Sample queries

9. Future Enhancements

- Multi-document support
- Feedback loop
- Memory integration
- Deployment

6. Final Submission Requirements

- HLD Document (PDF)
- LLD Document (PDF)
- Technical Documentation (PDF)
- Working Project (optional but preferred)

7. Evaluation Criteria

Criteria	Weight
HLD Quality	20%
LLD Depth	20%
Technical Documentation	25%
Concept Application	20%
Clarity & Presentation	15%

8. Important Instructions

- Do NOT copy generic architecture from the internet
- Design must reflect understanding of RAG + workflow systems
- Focus on clarity, reasoning, and structure
- Treat this like a real production system design document

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Final Brief to Students

You are not just building a chatbot.

You are designing a scalable AI system with decision-making capability.

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